(i) Laspeyre's quantity index

$$Q_{01}^{L} = \frac{\sum q_1 p_0}{\sum q_0 p_0} \times 100$$

$$= \frac{440}{266} \times 100$$

$$= 165.4$$

(ii) Paasche's quantity index

$$Q_{01}^{P} = \frac{\sum q_1 p_1}{\sum q_0 p_1} \times 100$$

$$= \frac{558}{342} \times 100$$

$$= 163.15$$

(iii) Fisher's quantity index

$$Q_{01}^{F} = \sqrt{Q_{01}^{L} \times Q_{01}^{P}}$$

$$= \sqrt{\frac{\sum q_1 p_0}{\sum q_0 p_0} \times \frac{\sum q_1 p_1}{\sum q_0 p_1}} \times 100$$

$$= \sqrt{\frac{440}{266} \times \frac{558}{342}} \times 100$$

$$= 1.6428 \times 100$$

$$= 164.28$$

6.4.4 Tests for Index numbers

Fisher has given some criteria that a good index number has to satisfy. They are called (i) Time reversal test (ii) Factor reversal test (iii) Circular test. Fisher has constructed in such a way that this index number satisfies all these tests and hence it is called Fisher's Ideal Index number.

Time reversal test

Fisher has pointed out that a formula for an index number should maintain time consistency by working both forward and backward with respect to time. This is called time reversal test. Fisher describes this test as follows.

"The test is that the formula for calculating an index number should be such that it gives the same ratio between one point of comparison and the other, no matter which of the two is taken as base or putting in another way the index number reckoned forward should be the reciprocal of that reckoned back ward". A good index number shoud satisfy the time reversal test.

This statement is expressed in the form of equation as $P_{01} \times P_{10} = 1$.

where

$$P_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}}$$

12th_Statistics_EM_Unit_6.indd 170 1/28/2020 6:37:35 PM